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Hedger of Last Resort: Evidence from Brazilian FX Interventions, Local Credit, and Global Financial Cycles

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1 Big picture

- **Question.** Can foreign-exchange interventions by an emerging-market central bank attenuate global financial spillovers to domestic credit and employment?
- **Policy studied.** The Central Bank of Brazil supplied FX derivatives after the 2013 Taper Tantrum, effectively selling insurance against exchange-rate risk and becoming a *hedger of last resort*.
- **Main mechanism.** Brazilian banks with more foreign debt were more exposed to global FX stress. When the real depreciated and volatility rose, these banks cut credit to local firms. Central-bank FX derivatives hedged this risk and relaxed the credit contraction.
- **Main empirical result.** After the Taper Tantrum, banks with higher ex ante foreign debt reduced credit supply. The FX intervention roughly halved the negative credit-supply effect.
- **Real effect.** Firms more exposed to affected banks experienced lower total credit and employment after the shock; these real effects were attenuated after the intervention.
- **Panel result.** A broader 2008Q1–2015Q2 panel using global FX shocks confirms that global financial-cycle shocks reduce local credit through banks’ foreign-debt exposure, and local FX interventions mitigate the effect.
- **Cost side.** The policy has fiscal costs and possible moral-hazard costs. The intervention especially expands credit through low-capital banks and to riskier firms.
- **Economic takeaway.** FX intervention can work as macro-financial insurance when private balance sheets are exposed to global dollar and FX shocks, but it transfers risk to the public sector and can distort risk-taking incentives.

2 Incremental contribution of the paper

- **From exchange rates to credit and employment.** Much FX-intervention evidence studies exchange-rate levels or volatility. This paper traces interventions through bank balance sheets to firm credit and employment.
- **Direct evidence on derivatives-based FX intervention.** The policy is not traditional spot reserve sales. The central bank provides FX swaps/derivatives, directly supplying hedges against FX risk.
- **Three matched administrative registers.** The paper links bank-firm credit, banks’ foreign credit, and employer-employee data, allowing the authors to connect global shocks to loan supply and firm-level real outcomes.
- **Bank exposure design.** The key treatment intensity is banks’ ex ante foreign debt. This identifies which domestic banks are more exposed to global FX stress.
- **Within-firm identification of credit supply.** Firm fixed effects and firm-quarter fixed effects compare credit changes across banks lending to the same firm, helping separate credit supply from firm credit demand.
- **Policy-offset design.** The paper measures whether central-bank derivatives interacted with bank FX exposure reverse the credit contraction caused by global shocks.
- **Costs of stabilization.** The paper adds a moral-hazard and fiscal-cost dimension to FX-intervention analysis, showing that hedging policy can increase risk-taking by weaker banks.

3 Data and measurement

3.1 Matched administrative data

The empirical analysis uses three administrative registers from Brazil:

- **bank-firm credit registry:** loan-level and bank-firm credit exposures;
- **foreign credit to banks:** banks' foreign debt and maturity structure;
- **employer-employee data:** firm employment, wages, and workforce characteristics.

Interpretation: The key advantage is that the paper can follow the entire transmission chain: global FX stress hits banks with foreign debt, banks adjust lending to firms, and firms adjust employment.

3.2 Treatment exposure: bank foreign debt

The central exposure variable is

$$\text{Bank FX Debt}_b = \frac{\text{foreign debt of bank } b}{\text{total liabilities of bank } b}.$$

It is measured before the shock or policy window, so it is predetermined relative to the outcome period.

Interpretation: A bank with higher foreign debt has larger rollover and balance-sheet exposure to global financial tightening and currency depreciation. This is the cross-sectional source of shock exposure.

3.3 Global financial shocks

The paper focuses on the 2013 Taper Tantrum and a broader panel of global FX shocks. The Taper Tantrum is a sharp external shock to emerging-market financial conditions. In the panel, global shocks are measured with an emerging-market exchange-rate index and related capital-flow and global-financial-cycle measures.

Interpretation: The global shock is plausibly external to a single Brazilian firm or bank. The identification comes from comparing banks with different pre-existing FX exposure when the global shock arrives.

3.4 FX intervention

The Central Bank of Brazil supplied FX derivatives after the Taper Tantrum. The main policy indicator is denoted

$$\text{FX Intv}_t.$$

It equals one during periods of active central-bank FX intervention or is measured as the scale of FX derivatives supplied by the central bank in the broader panel.

Interpretation: The policy does not mainly work through direct credit to firms. It works by insuring banks' FX risk, which changes their willingness or capacity to supply domestic credit.

3.5 Outcomes

The paper studies:

- bank-firm credit growth;
- firm total credit growth;
- firm employment, wages, and tenure;
- bond/loan-level and bank controls;
- moral-hazard outcomes through interactions with bank capital and firm risk proxies.

Interpretation: The outcomes distinguish financial transmission from real effects and from risk-taking costs.

4 Empirical specifications

4.1 Taper Tantrum credit-supply specification

The baseline post-Taper regression at the firm-bank level is:

$$\Delta \text{Credit}_{f,b} = \beta_1 \text{Bank FX Debt}_b + \gamma X_{f,b} + \theta_f + \varepsilon_{f,b}.$$

Here $\Delta \text{Credit}_{f,b}$ is the log growth rate of credit from bank b to firm f during the post-Taper window, and θ_f is a firm fixed effect.

Interpretation: With firm fixed effects, the estimate compares different banks lending to the same firm. This is a credit-supply comparison: if exposed banks cut more to the same firm, the effect is not simply lower demand by that firm.

4.2 Taper plus intervention specification

To test the effect of the intervention, the paper estimates:

$$\Delta \text{Credit}_{f,b,t} = \beta_1 \text{Bank FX Debt}_b + \beta_2 (\text{FX Intv}_t \times \text{Bank FX Debt}_b) + \gamma X_{f,b} + \theta_{f,t} + \varepsilon_{f,b,t}.$$

Interpretation: A negative β_1 means foreign-debt-exposed banks cut credit after the shock. A positive β_2 means the FX intervention mitigates that credit contraction.

4.3 Firm-level real-effects specification

The firm-level specification aggregates bank exposures to the firm level:

$$\Delta Y_f = \beta \widehat{\text{Exposure}}_f + \Gamma Z_f + \varepsilon_f,$$

where Y_f includes total credit, employment, wages, and tenure. Exposure reflects the firm's borrowing from more FX-exposed banks.

Interpretation: This step asks whether bank credit-supply shocks matter for firms, not only for bank-firm loan records.

4.4 Full-panel global-shock specification

In the 2008Q1–2015Q2 panel, the paper estimates versions of:

$$\begin{aligned} \Delta \text{Credit}_{f,b,t} = & \beta_1 (\text{Bank FX Debt}_{b,t-1} \times \Delta \text{EMEFX}_t) \\ & + \beta_2 (\text{Bank FX Debt}_{b,t-1} \times \Delta \text{EMEFX}_t \times \text{FX Intv}_t) + FE + Controls + \varepsilon_{f,b,t}. \end{aligned}$$

Interpretation: This generalizes the Taper Tantrum evidence. The question is whether global FX shocks repeatedly reduce credit supply for foreign-debt-exposed banks and whether FX interventions repeatedly offset the shock.

4.5 Risk-taking specification

The moral-hazard test adds bank capital and firm-risk interactions:

$$\Delta\text{Credit}_{f,b,t} = \beta_1(\text{Bank FX Debt}_b \times \text{FX Intv}_t) + \beta_2(\text{Bank FX Debt}_b \times \text{Capital}_b \times \text{FX Intv}_t) \\ + \beta_3(\text{Capital}_b \times \text{FX Intv}_t) + FE + Controls + \varepsilon_{f,b,t}.$$

Interpretation: If the intervention encourages more risk-taking by weak banks, the credit expansion should be stronger for less-capitalized banks and riskier firms.

5 Identification and empirical design

5.1 Core identifying variation

The design combines three ingredients:

1. an external global financial shock, the Taper Tantrum or global FX shocks;
2. pre-existing bank exposure to foreign debt;
3. central-bank FX derivatives that vary over time.

Interpretation: The treatment is not bank foreign debt alone. It is foreign debt exposure interacted with global FX stress, and then with central-bank intervention.

5.2 Why firm fixed effects matter

Firm fixed effects compare the credit extended by different banks to the same firm:

$$\Delta\text{Credit}_{f,b} = \beta\text{Bank FX Debt}_b + \theta_f + \varepsilon_{f,b}.$$

Interpretation: This is crucial because firms may have different credit demand. If the same firm receives less credit from the more FX-exposed bank, the result is more plausibly credit supply.

5.3 Why firm-quarter fixed effects matter

In panel specifications, firm-quarter fixed effects absorb time-varying firm demand:

$$\theta_{f,t}.$$

Interpretation: This means the coefficient is identified from how different banks change credit to the same firm in the same quarter when global FX shocks and interventions occur.

5.4 Policy-offset interpretation

The interaction coefficient on Bank FX Debt $\times$ FX Intv asks whether the central bank's hedging policy offsets the negative effect of bank foreign-debt exposure.

Interpretation: A positive interaction does not merely show that all credit rose during interventions. It shows that the banks most hurt by FX exposure contract less after the policy.

5.5 Threats and responses

- **Bank selection.** Banks with more foreign debt may differ systematically. The paper controls for size, capital, nonperforming loans, state ownership, FX exposure, trade exposure, and maturity structure.
- **Firm demand.** Firm and firm-quarter fixed effects absorb borrower demand shocks.
- **Macro shocks.** Panel tests use global FX shocks, capital-flow shocks, and macro controls.
- **Other policy tools.** Robustness checks account for monetary policy, reserve requirements, and macro-prudential measures.
- **Moral hazard.** Table VI directly tests whether low-capital banks and risky firms receive more credit after the intervention.

Interpretation: The identification is strongest when all pieces line up: exposed banks cut credit after global FX stress; the same exposure is offset during intervention; firms experience real effects; and risk-taking costs appear in the predicted places.

6 Tables and figures

6.1 Table I: QE Tapering: Credit Supply, Firm-Bank Level; Table II: QE Tapering versus FX Interventions: Credit Supply, Firm-Bank Level

Read: Table I estimates the immediate post-Taper effect. The coefficient on Bank FX Debt is negative and significant across specifications. Table II adds the FX intervention interaction. The interaction coefficient is positive and significant, offsetting roughly half the negative effect.

Economic message: The Taper Tantrum tightened credit supply through banks with foreign debt, and the central-bank derivatives program mitigated the contraction.

Table I is the baseline credit-supply result. Table II is the policy result. The comparison is the paper's central narrative: global FX stress hits foreign-debt-exposed banks, then BCB hedging weakens the pass-through to firms.

Table I
QE Tapering: Credit Supply, Firm-Bank Level

The table reports estimates of the equation

$$\Delta Credit_{f,b} = \beta_1 Bank\ FX\ Debt_b + \gamma X_{f,b} + \theta_f + \epsilon_{f,b},$$

where $\Delta Credit_{f,b}$ is log growth rate of credit provided to firm f by bank b , over one quarter after Tapering Speech (end of April '13–end of July '13), $Bank\ FX\ Debt_b$ is bank's ex ante share of foreign debt in its total liabilities, θ_f is firm fixed effect, and $X_{f,b}$ is a set of controls; all explanatory variables are measured as of the end of April '13. In column (7), credit growth rate is calculated as the net flow of credit provided by each bank to each firm during the quarter relative to the average credit over the period. Constant in column (1) is omitted. Weights in column (6) are proportional to (log of) firm employment. Standard errors (in parenthesis) are calculated under two-way clustering by bank and firm industry (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$).

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>Bank FX Debt</i>	-1.06*** (0.26)	-0.92*** (0.20)	-1.07*** (0.19)	-0.77*** (0.22)	-0.89*** (0.24)	-0.83*** (0.27)	-1.20*** (0.41)
<i>Loan-level controls:</i>							
<i>Unused Credit Line</i>			0.16*** (0.01)	0.14*** (0.01)	0.13*** (0.01)	0.13*** (0.01)	-0.43*** (0.02)
<i>Default</i>			-0.05*** (0.01)	-0.04*** (0.01)	-0.04*** (0.01)	-0.04*** (0.01)	-0.24*** (0.02)
<i>Bank Share</i>			-0.03 (0.02)	-0.08*** (0.02)	-0.08*** (0.02)	-0.07*** (0.02)	-0.54*** (0.04)
<i>Bank-level controls:</i>							
<i>Size</i>				0.03*** (0.00)	0.02*** (0.00)	0.02*** (0.00)	0.05*** (0.01)
<i>Capital</i>				0.49** (0.18)	0.46*** (0.11)	0.49*** (0.13)	1.33*** (0.36)
<i>NPL</i>				0.13 (0.17)	-0.08 (0.10)	-0.17 (0.12)	-1.21*** (0.26)
<i>FX Debt in Loans</i>				0.04*** (0.01)	0.08*** (0.01)	0.06*** (0.02)	0.15*** (0.02)
<i>State owned</i>				0.03 (0.02)	0.00 (0.02)	0.01 (0.02)	0.03 (0.04)
<i>Exposure to Trade</i>				0.16** (0.08)	0.17*** (0.06)	0.16** (0.06)	0.15 (0.14)
<i>Net FX Exposure</i>				3.54** (1.44)	1.16 (1.00)	1.46 (1.16)	3.17 (3.10)
<i>FX Debt < 1y</i>				-0.17*** (0.03)	-0.16*** (0.04)	-0.50*** (0.07)	
<i>FX Debt > 3y</i>				0.03 (0.03)	0.02 (0.03)	-0.01 (0.05)	
<i>Firm FE</i>	no	yes	yes	yes	yes	yes	yes
<i>WLS</i>	no	no	no	no	no	yes	yes
<i>R²</i>	0.01	0.41	0.42	0.43	0.43	0.41	0.45
<i>N</i>	114,182	114,182	114,182	114,182	114,182	114,182	131,077

Table I: Taper Tantrum credit-supply contraction at the firm-bank level.

Tables I and II are paired because the first establishes the shock and the second measures the policy offset.

6.2 Table III: QE Tapering versus FX Interventions: Firm-Level Evidence; Table IV: GFC Shocks, Full Panel

Read: Table III aggregates the effect to firms. Credit and employment decline for firms exposed to foreign-debt-dependent banks, and the intervention weakens the effect. Table IV expands the design to a full panel of global financial-cycle shocks.

Economic message: The effects are not confined to loan records; they transmit to total firm credit and employment. The panel evidence shows that the mechanism is broader than one episode.

Table III gives the real-effects evidence. Table IV shows that the same credit-supply channel is present during other global FX shock episodes in 2008–2015.

Table II
QE Tapering versus FX Interventions: Credit Supply, Firm-Bank Level

The table reports estimates of the equation

$$\Delta Credit_{f,b,t} = \beta_1 Bank\ FX\ Debt_b + \beta_2 FX\ Intv(0/1)_t \times Bank\ FX\ Debt_b + \gamma X_{f,b} + \theta_{f,t} + \epsilon_{f,b,t},$$

where $\Delta Credit_{f,b,t}$ is quarterly log growth rate of credit provided to firm f by bank b , $Bank\ FX\ Debt_b$ is bank's ex ante share of foreign debt in its total liabilities, $\theta_{f,t}$ is firm-quarter fixed effect, and $X_{f,b}$ is a set of controls. $FX\ Intv_t$ is equal to one for periods t of active BCB FX intervention program, and zero otherwise. The left panel spans the period of end of April '13–end of October '13 (two quarters with one quarter of the postpolicy period). The right panel spans the period of end of April '13–end of April '14 (four quarters with three quarters of the postpolicy period). Constants in columns (1) and (5) are omitted. The set of control variables is identical to that in Table I. All explanatory variables are measured as of the end of April '13. Weights in columns (4) and (8) are proportional to (log of) firm employment. Standard errors (in parenthesis) are calculated under two-way clustering by bank and firm industry (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$).

	+1 Policy Quarter				+3 Policy Quarters			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Bank FX Debt</i>	-1.06*** (0.26)	-0.92*** (0.20)	-1.14*** (0.23)	-1.13*** (0.26)	-1.06*** (0.26)	-0.92*** (0.20)	-1.18*** (0.20)	-1.19*** (0.24)
<i>Bank FX Debt × FX Intv (0/1)</i>	0.48*** (0.22)	0.42*** (0.16)	0.44*** (0.16)	0.40*** (0.18)	0.57*** (0.16)	0.51*** (0.11)	0.52*** (0.12)	0.47*** (0.14)
<i>FX Intv (0/1)</i>	-0.03** (0.01)				-0.04*** (0.01)			
<i>Firm × Time FE</i>	no	yes	yes	yes	no	yes	yes	yes
<i>Controls</i>	no	no	yes	yes	no	no	yes	yes
<i>WLS</i>	no	no	no	yes	no	no	no	yes
<i>R²</i>	0.01	0.41	0.43	0.41	0.004	0.41	0.43	0.41
<i>N</i>	229,327	229,327	229,327	229,327	462,677	462,677	462,677	462,677

Table II: FX intervention offsets the bank FX-debt shock.

Table III
QE Tapering versus FX Interventions: Firm-Level Evidence
The table reports estimates of the equation

$$\Delta Y_{f,t} = \beta_1 \text{Bank FX Debt}_{f,t} + \beta_2 \text{FX Intv}_{f,t}(0/1) + \beta_3 \text{Bank FX Debt}_{f,t} + \gamma X_{f,t} + \theta_{f,t} + \epsilon_{f,t}$$

where $\Delta Y_{f,t}$ is either the log growth rate of total credit of a firm f from all banks included in the sample, columns (1) to (2), or from all credit institutions, columns (3) to (4), or the log growth rate of firm employment, or the change in the average log of tenure of firm's employees, *Bank FX Debt_{f,t}* is a weighted average of firm lenders' ex ante share of foreign debt in their total liabilities, $\theta_{f,t}$ is industry-location-time fixed effect, and $X_{f,t}$ is a set of controls. The set of control variables include those reported in Table I as well as *log of Total Credit*, *log of Total Employment*, *Average log of Wage*, and *Average log of Tenure*. Bank-firm and bank-level variables are aggregated to the firm level by taking the weighted average of the corresponding variables with weights proportional to the bank's share in firm total ex ante credit liabilities. All explanatory variables are measured as of the end of April13. *Proxy for Credit Demand* refers to estimated firm fixed effect from the firm-bank regression. Weights in columns (2), (4), (6), and (8) are proportional to log of firm employment. Standard errors (in parenthesis) are calculated under two-way clustering by main bank and firm industry (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$).

	ΔCredit		$\Delta \text{Total Credit}$		$\Delta \text{Employment}$		ΔTenure	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Bank FX Debt</i>	-1.11*** (0.20)	-1.17*** (0.19)	-0.80*** (0.22)	-0.83*** (0.21)	-0.11** (0.07)	-0.20*** (0.07)	0.23** (0.12)	0.25** (0.11)
<i>Bank FX Debt × FX Intv (0/1)</i>	0.78*** (0.07)	0.77*** (0.08)	0.67*** (0.10)	0.65*** (0.10)	0.10** (0.05)	0.11** (0.05)	-0.13** (0.08)	-0.14** (0.08)
<i>Proxy for Credit Demand</i>	0.90*** (0.01)	0.90*** (0.01)	0.77*** (0.01)	0.75*** (0.01)	0.02*** (0.00)	0.02*** (0.00)	-0.02*** (0.00)	-0.02*** (0.00)
Industry × Location × Time FE	yes	yes	yes	yes	yes	yes	yes	yes
Controls	yes	yes	yes	yes	yes	yes	yes	yes
WLS	no	no	no	no	yes	no	yes	yes
R ²	0.80	0.79	0.67	0.66	0.13	0.15	0.14	0.15
N	180,679	180,679	180,679	180,679	180,679	180,679	180,679	180,679

Table III: firm-level credit and employment evidence.

These tables connect the event design to real firm outcomes and to broader panel evidence.

6.3 Table V: GFC Shocks and FX Interventions, Full Panel; Table VI: QE Tapering versus FX Interventions: Risk Taking

Read: Table V shows that FX interventions also mitigate global financial-cycle shocks in the full panel. Table VI studies costs: less-capitalized banks expand credit more after the intervention, and the effects are stronger for riskier firms.

Economic message: FX intervention is effective but not free. It stabilizes credit, but by insuring bank balance sheets it can encourage risk-taking through weak banks.

Pairing Tables V and VI highlights the policy trade-off. Table V is the stabilization benefit; Table VI is the moral-hazard cost.

Table IV
GFC Shocks, Full Panel

The table reports estimates of the equation

$$\Delta \text{Credit}_{f,b,t} = \beta_1 \text{Bank FX Debt}_{b,t-1} + \beta_2 \text{Bank FX Debt}_{b,t-1} \times \Delta \text{EME FX}_{t-1} + \gamma X_{f,b,t-1} + \theta_{f,t} + \epsilon_{f,b,t}$$

where $\Delta \text{Credit}_{f,b,t}$ is quarterly log growth rate of credit provided to firm f by bank b , *Bank FX Debt_{b,t}* is bank's share of foreign debt in its total liabilities, $\theta_{f,t}$ is firm-quarter fixed effect, and $X_{f,b,t}$ is a set of controls. The sample period is 2008Q1 to 2015Q2. In all columns, the estimates are conditioned on lagged bank- and bank-level control variables (the set of control variables is identical to that in Table I). *Capital Flows (Quantity)* stands for change in log of end-of-quarter levels of aggregate external debt of the banking sector in Brazil. *Capital Flows (Price)* stands for change in interest payments on external debt to be paid during the following year normalized by the end-of-quarter levels of aggregate external debt of the banking sector in Brazil. Additional macroeconomic variables interacted with *Bank FX Debt* in columns (3) and (7) include changes in *BRA Money Rate*, *Inflation*, *IBC BR*, and *VIX*. Standard errors (in parenthesis) are calculated under two-way clustering by bank and quarter-industry (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$).

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Bank FX Debt × ΔEME FX Index</i>	-3.34*** (1.11)	-3.34*** (0.96)	-8.50*** (2.57)	-3.73*** (0.94)				
<i>Bank FX Debt × ΔEME FX Vol</i>					-0.36** (0.18)	-0.70*** (0.18)	-1.89*** (0.58)	-0.71*** (0.18)
<i>Bank FX Debt × ΔUS Shadow Rate</i>		-0.27*** (0.07)	-0.05 (0.11)	-0.24*** (0.08)		-0.38*** (0.08)	-0.37*** (0.08)	-0.37*** (0.07)
<i>Bank FX Debt × Δ Capital Flows (Quantity)</i>				-0.37 (0.40)				-0.01 (0.43)
<i>Bank FX Debt × Δ Capital Flows (Price)</i>				0.08 (0.08)				0.05 (0.08)
Firm × Quarter FE	yes	yes	yes	yes	yes	yes	yes	yes
Controls	yes	yes	yes	yes	yes	yes	yes	yes
Macro interactions	no	no	yes	no	no	no	yes	no
R ²	0.43	0.43	0.43	0.43	0.43	0.43	0.43	0.43
N	3,900,653	3,900,653	3,900,653	3,900,653	3,900,653	3,900,653	3,900,653	3,900,653

Table IV: full-panel global financial-cycle shocks.

Table V
GFC Shocks and FX Interventions, Full Panel
The table reports estimates of the equation

$$\Delta \text{Credit}_{f,b,t} = \beta_1 \text{Bank FX Debt}_{b,t-1} + \beta_2 \text{Bank FX Debt}_{b,t-1} \times \Delta \text{EME FX}_{t-1} + \beta_3 \text{Bank FX Debt}_{b,t-1} \times \text{FX Intv}_{f,t}(0/1) + \beta_4 \text{Bank FX Debt}_{b,t-1} \times \Delta \text{EME FX}_{t-1} \times \text{FX Intv}_{f,t}(0/1) + \gamma X_{f,b,t-1} + \theta_{f,t} + \epsilon_{f,b,t}$$

where $\Delta \text{Credit}_{f,b,t}$ is quarterly log growth rate of credit provided to firm f by bank b , *Bank FX Debt_{b,t}* is bank's share of foreign debt in its total liabilities, $\theta_{f,t}$ is firm-quarter fixed effect, and $X_{f,b,t}$ is a set of controls. *FX Intv_{f,t}(0/1)* is equal to the end-of-quarter notional amount of BCB swap intervention program over FX reserves. The sample period is 2008Q1 to 2015Q2. In all columns, the estimates are conditioned on the interaction of lagged *Bank FX Debt* and changes in the *US Shadow Rate*, as well as on lagged bank- and bank-level control variables (the set of control variables is identical to that in Table I). Additional macroeconomic variables interacted with *Bank FX Debt* include: changes in *BRA Money Rate*, *Inflation*, *IBC BR*, and *VIX*. In columns (2), (4), (6), and (8), *Bank FX Debt* is additionally interacted with *Capital Flows (Quantity)* and *Capital Flows (Price)*. Standard errors (in parenthesis) are calculated under two-way clustering by bank and quarter-industry (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$).

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Bank FX Debt × ΔEME FX Index</i>	-8.40*** (2.11)	-8.79*** (2.36)	-11.96*** (3.33)	-12.22*** (3.88)				
<i>Bank FX Debt × ΔEME FX Index × FX Intv</i>	25.07*** (10.02)	25.45*** (10.34)	23.35*** (7.26)	22.40*** (6.99)				
<i>Bank FX Debt × ΔEME FX Vol</i>					-1.30*** (0.35)	-1.29*** (0.35)	-2.50*** (0.73)	-3.49*** (1.07)
<i>Bank FX Debt × ΔEME FX Vol × FX Intv</i>					5.65*** (2.97)	5.66*** (2.99)	7.55*** (2.21)	9.51*** (2.64)
Firm × Quarter FE	yes	yes	yes	yes	yes	yes	yes	yes
Controls	yes	yes	yes	yes	yes	yes	yes	yes
Macro interactions	no	no	yes	no	no	yes	yes	yes
Interactions with capital flows	no	no	no	yes	no	yes	no	yes
R ²	0.43	0.43	0.43	0.43	0.43	0.43	0.43	0.43
N	3,900,653	3,900,653	3,900,653	3,900,653	3,900,653	3,900,653	3,900,653	3,900,653

Table V: intervention offsets global FX shocks in the full panel.

The policy trade-off: stabilization benefit versus risk-taking costs.

Table VI
QE Tapering versus FX Interventions: Risk Taking
The table reports estimates of versions of the equation

$$\Delta \text{Credit}_{f,b,t} = \beta_1 \text{Bank FX Debt}_{b,t} \times \text{FX Intv}_{f,t}(0/1) + \beta_2 \text{Bank FX Debt}_{b,t} \times \text{Capital}_{b,t} \times \text{FX Intv}_{f,t}(0/1) + \beta_3 \text{Capital}_{b,t} \times \text{FX Intv}_{f,t}(0/1) + \gamma X_{f,b,t} + \theta_{f,t} + \phi_b + \epsilon_{f,b,t}$$

where $\Delta \text{Credit}_{f,b,t}$ is quarterly log growth rate of credit provided to firm f by bank b , *Bank FX Debt_{b,t}* is bank's ex ante share of foreign debt in its total liabilities, *Capital_{b,t}* is bank's ex ante capital, $\theta_{f,t}$ is firm-quarter fixed effect, ϕ_b is bank fixed effect, and $X_{f,b,t}$ is a set of controls. *FX Intv_{f,t}(0/1)* is equal to one for the period t of active BCB FX intervention program, and zero otherwise. All regressions are estimated using the end of April '13–end of October '13 sample (two quarters with one quarter of the postpolicy period). Controls include the following variables: *Unused Credit Line*, *Default*, and *Bank Share*. All explanatory variables are measured as of the end of April '13. Columns (2) and (3) report the estimates obtained on a subsample of firms with ex ante average interest rates below the sample median (column (2)) and above the sample median (column (3)). Columns (4) and (5) report the estimates obtained on a subsample of firms with greater than 10 employees (column (4)) and not more than 10 employees (column (5)). Standard errors (in parenthesis) are calculated under two-way clustering by bank and firm industry (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$).

	All Firms (1)	Interest Rate (2) (3)		Firm Size (4) (5)	
		Low	High	Large	Small
<i>Bank FX Debt × FX Intv (0/1)</i>	0.32*** (0.10)	0.26** (0.11)	0.30** (0.13)	0.41*** (0.11)	0.03 (0.13)
<i>Bank FX Debt × Capital × FX Intv (0/1)</i>	-9.71*** (3.01)	-3.31 (2.04)	-16.18*** (3.65)	-6.52** (3.30)	-16.72*** (2.49)
<i>Capital × FX Intv (0/1)</i>	0.03 (0.08)	-0.04 (0.08)	0.12 (0.12)	0.03 (0.10)	0.17 (0.12)
Firm × Time FE	yes	yes	yes	yes	yes
Bank FE	yes	yes	yes	yes	yes
Controls	yes	yes	yes	yes	yes
R ²	0.43	0.43	0.43	0.41	0.46
N	229,327	113,746	115,580	129,242	100,080

Table VI: risk-taking channel and low-capital banks.

6.4 Figure A.1: Bank FX Debt, the BRL/USD Exchange Rate, and the Intervention; Figure A.2: Maturity Composition of Foreign Debt and FX Derivatives

Read: Figure A.1 shows the sequence of the Taper speech, BRL depreciation/volatility, and the BCB derivatives position. Figure A.2 shows the maturity structure of foreign debt and FX derivatives.

Economic message: The figures provide the macro and balance-sheet context: banks faced FX rollover and valuation stress, and the central bank supplied derivatives to absorb that risk.

Figure A.1 visualizes why the Taper Tantrum is a relevant global shock for Brazilian banks. Figure A.2 helps interpret the mechanism by showing the short-term structure of FX liabilities and hedging instruments.

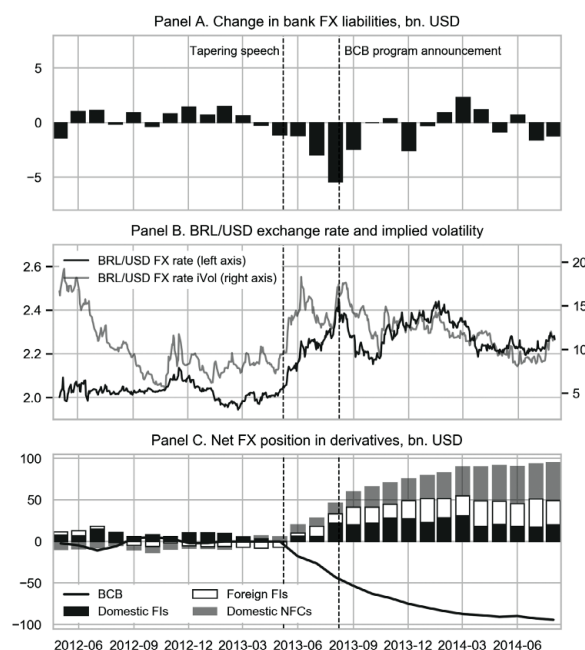


Figure A.1. Bank FX Debt, the BRL/USD exchange rate, and the intervention by the Central Bank of Brazil. Changes in the bank FX liabilities (upper panel, monthly series) are calculated based on the register of foreign claims on domestic financial institutions (ROF). Net FX positions in derivatives (lower panel, from Brazil stock exchange, monthly series) are aggregated by sector over all instruments with FX exposure, including BCB swaps. Vertical grid lines indicate middle-of-month dates. Net FX positions are end-of-month.

Figure A.1: FX debt, exchange rate, volatility, and BCB intervention.

Appendix figures A.1 and A.2 show the institutional timing and balance-sheet channel.

6.5 Figure A.3: Correlation between the BRL/USD Exchange Rate and EME FX Index

Read: Figure A.3 plots the correlation between BRL/USD movements and the EME FX index used in the panel specifications.

Economic message: The positive association supports the use of external EME FX shocks as a proxy for global FX pressure relevant to Brazil.

This figure supports the panel strategy. If the EME FX index were unrelated to Brazil's FX stress, the global-shock interpretation would be weaker.

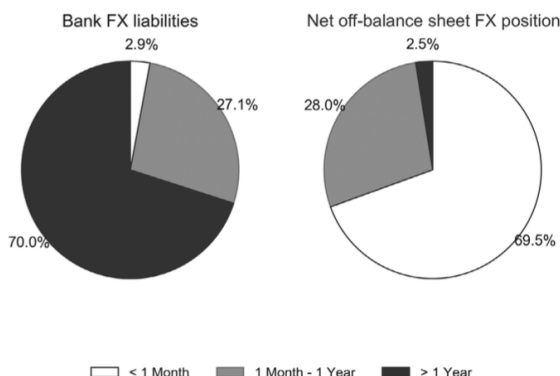


Figure A.2. Maturity composition of foreign debt and FX derivatives. The maturity structure of bank FX liabilities is based on the register of foreign claims on domestic financial institutions (ROF). The maturity structure of off-balance-sheet FX positions is based on *Demonstrativo de Risco de Mercado* (DRM). DRM is a regulatory form filed by all financial institutions in Brazil, that provides details about underlying market risk factors to the Bank Supervision of the BCB. Off-balance FX positions are net long exposures. The graphs reflect the aggregate values for the banks in our sample in April 2013.

Figure A.2: maturity composition of foreign debt and FX derivatives.

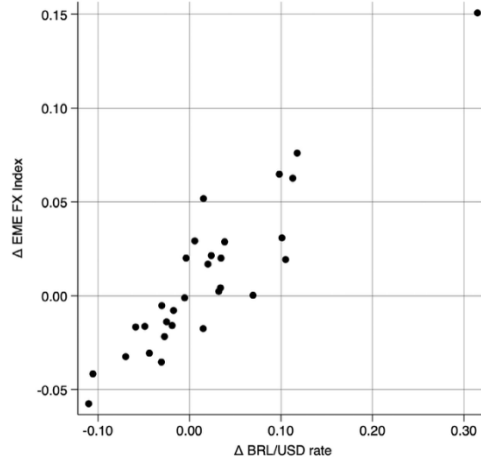


Figure A.3. Correlation between the BRL/USD exchange rate and EME FX index. Quarterly log growth rates, 2008Q1 to 2015Q2.

Figure A.3: Correlation between the BRL/USD exchange rate and EME FX index.

7 Short summary

- The paper studies whether FX interventions can reduce global financial spillovers to domestic credit and employment.
- Brazil's 2013 Taper Tantrum episode provides a sharp global shock and a large FX-derivative intervention.
- Banks with more foreign debt cut credit more after the shock.
- The BCB intervention supplied FX insurance and roughly halved the credit contraction.
- Firm-level evidence shows that affected firms experienced lower total credit and employment, and the intervention mitigated those effects.
- Panel evidence from 2008Q1–2015Q2 confirms the mechanism for broader global FX shocks.
- The policy works through a hedging channel: it transfers FX risk from banks to the central bank.
- The policy has costs, including fiscal exposure and moral-hazard risk-taking by weaker banks.

8 Conclusion

8.1 Core conclusion

Barbone Gonzalez, Khametshin, Peydro, and Polo show that FX intervention can be effective when it works as balance-sheet insurance. The Brazilian central bank's derivative intervention reduced the credit-supply contraction generated by global FX stress and weakened the real effect on firms.

8.2 Economic intuition

Banks with foreign debt are vulnerable to global dollar tightening and local currency depreciation. When funding conditions deteriorate, they cut domestic credit. FX derivatives from the central bank hedge this balance-sheet risk and therefore relax the incentive or need to contract credit. This is why the policy acts as a hedger of last resort.

8.3 Incremental contribution revisited

The paper’s main contribution is to connect FX intervention to microeconomic credit and employment outcomes. It moves beyond exchange-rate effects and shows that policy can stabilize the domestic credit channel. The matched administrative data allow the authors to observe the entire chain from foreign-bank liabilities to firm-level employment.

8.4 Policy implications

FX intervention can be a useful complement to monetary, macroprudential, and capital-flow policies in emerging markets. However, because the central bank assumes risk, the intervention has fiscal costs. It may also induce moral hazard when weakly capitalized banks expand credit to risky firms after receiving public-sector hedging.

8.5 Limitations

The setting is Brazil, a large emerging market with rich data and a particular derivative-intervention technology. Effects may differ in countries with different bank balance sheets, foreign-currency liabilities, central-bank credibility, or fiscal capacity. The policy also depends on the central bank’s ability to manage derivative exposures.

8.6 Takeaway

The key lesson is that FX intervention is not only an exchange-rate policy. When implemented through derivatives, it is a risk-transfer policy. It can stabilize local credit during global financial shocks, but it must be evaluated jointly with its fiscal and moral-hazard costs.